In [1]:
```
!pip install pandas matplotlib scikit-learn
```

```
Requirement already satisfied: pandas in /opt/conda/lib/python3.11/site-packages
(3.0.6)
Requirement already satisfied: matplotlib in /opt/conda/lib/python3.11/site-packa
ges (3.11.2)
Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.11/site-pac
kages (1.9.1)
Requirement already satisfied: numpy>=1.26.0 in /opt/conda/lib/python3.11/site-pa
ckages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.1
1/site-packages (from pandas) (2.8.2)
Requirement already satisfied: contourpy>=1.0.1 in /opt/conda/lib/python3.11/site
-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /opt/conda/lib/python3.11/site-pac
kages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in /opt/conda/lib/python3.11/sit
e-packages (from matplotlib) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /opt/conda/lib/python3.11/sit
e-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.11/site-
packages (from matplotlib) (23.2)
Requirement already satisfied: pillow>=9 in /opt/conda/lib/python3.11/site-packag
es (from matplotlib) (13.0.0)
Requirement already satisfied: pyparsing>=3 in /opt/conda/lib/python3.11/site-pac
kages (from matplotlib) (3.3.3)
Requirement already satisfied: scipy>=1.10.0 in /opt/conda/lib/python3.11/site-pa
ckages (from scikit-learn) (1.17.1)
Requirement already satisfied: joblib>=1.4.0 in /opt/conda/lib/python3.11/site-pa
ckages (from scikit-learn) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in /opt/conda/lib/python3.11/site-
packages (from scikit-learn) (2.27.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in /opt/conda/lib/python3.11/
site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in /opt/conda/lib/python3.11/site
-packages (from joblib>=1.4.0->scikit-learn) (3.1.2)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.11/site-package
s (from python-dateutil>=2.8.2->pandas) (1.16.0)
```

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score,f1_score, classification_report


#load the data
df = pd.read_csv('/tmp/airline-weather.csv')

#select input and output columns
features = df[['Weather','CrewAvailable','BusyRunways']].copy()
target = df['DelayedYN'].astype(str).str.strip().str.lower()
target = target.map({'no': 0, 'yes': 1})


#convert text features into numbers
for column in features.columns:
  features[column] = features[column].astype('category').cat.codes
```

```
features_train, features_test, target_train, target_test = train_test_split(
    features, target, test_size=0.2, random_state=42
)

#train the model
model = RandomForestClassifier(random_state=42)
model.fit(features_train, target_train)

#predict the test data
predictions = model.predict(features_test)

#display model performance
print("Accuracy:", accuracy_score(target_test, predictions))
print("F1 Score:", f1_score(target_test, predictions))
print("\n Classification Report:")
print(classification_report(target_test,predictions))

#display feature importance
importance = pd.Series(
    model.feature_importances_,
    index=features.columns
)

print("\n Features Importance:")
print(importance.sort_values(ascending=False))

#Plot features importance
importance.sort_values().plot(kind='barh')
plt.title('Feature Importance')
plt.xlabel('Importance')
plt.ylabel('Features')
plt.tight_layout()
plt.show()
```

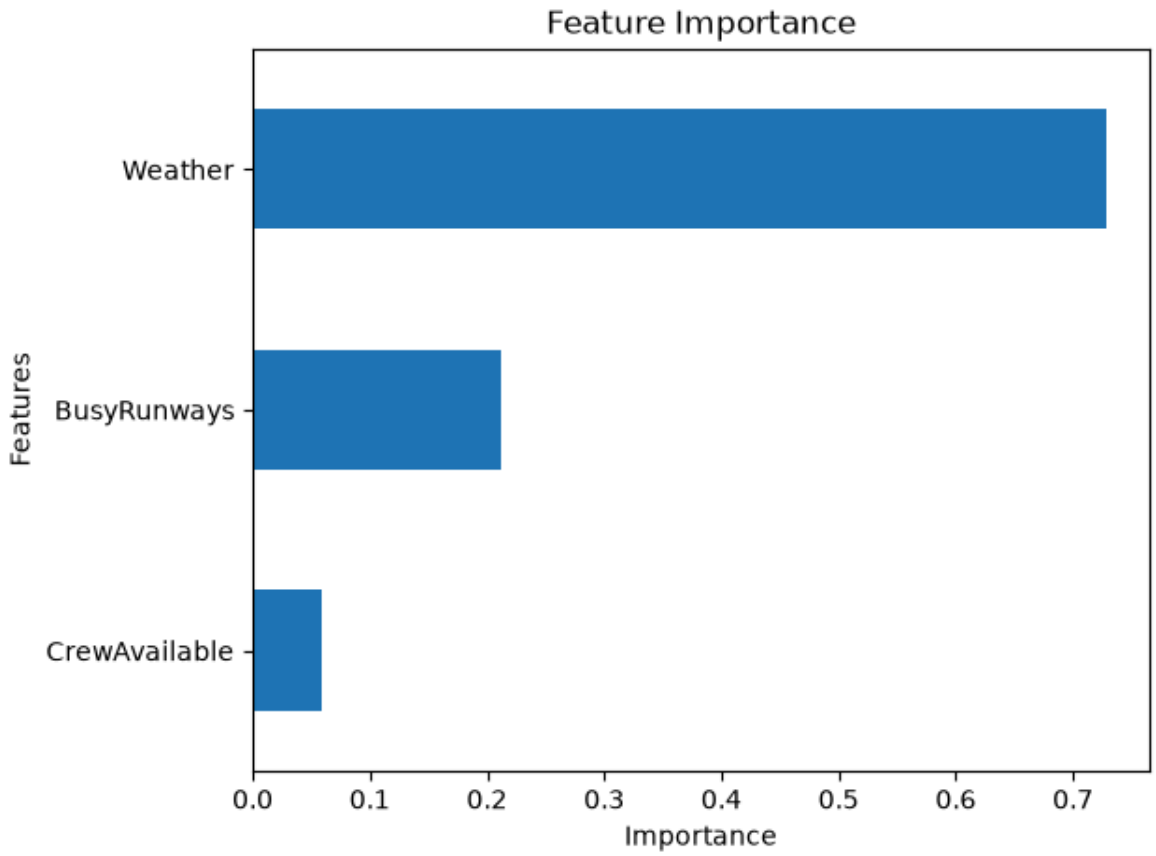
```
Accuracy: 0.7777777777777778
F1 Score: 0.8461538461538461

 Classification Report:
              precision    recall  f1-score   support

           0       0.60      0.60      0.60         5
           1       0.85      0.85      0.85        13

    accuracy                           0.78        18
   macro avg       0.72      0.72      0.72        18
weighted avg       0.78      0.78      0.78        18


 Features Importance:
Weather          0.728665
BusyRunways      0.212617
CrewAvailable    0.058718
dtype: float64
```


Feature Importance
Weather
BusyRunways
CrewAvailable
Features
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Importance

In [ ]: